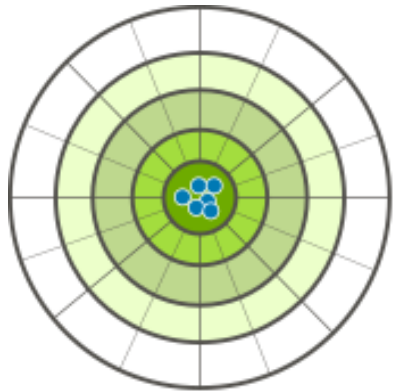


화학수학

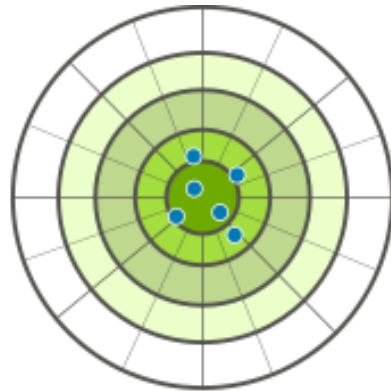
스무 번째 수업

확률과 통계

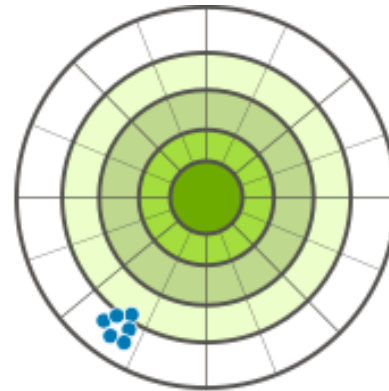
Precision (정밀도) and Accuracy (정확도)



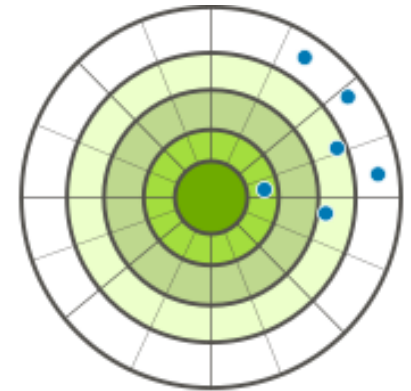
accurate
precise



accurate
not precise



not accurate
precise



not accurate
not precise

random errors

systematic errors

Probability Theory

study of random events

- **Classical systems:** unknown information
- **Quantum systems:** intrinsic randomness
- **Population:** large set of objects or numbers
 - Classical systems: imaginary set of infinitely many repetitions of a given measurement
 - Quantum systems: imaginary set of infinitely many replicas of the physical system

Probability Distribution (확률분포)

mathematical function that gives
the probabilities of occurrence of
different possible outcomes for an experiment

- Discrete probability distribution (이산 확률분포):
possible outcomes have discrete values
- Continuous probability distribution (연속 확률분포):
possible outcomes have continuous values

Probability Distributions

	Discrete	Continuous
Variable	$x_1, x_2, \dots, x_n$	x : real in $[x_{\min}, x_{\max}]$
Probability distribution	p_i	$f(x) dx$
Normalization	$\sum_{i=1}^n p_i = 1$	$\int_{x_{\min}}^{x_{\max}} f(x) dx = 1$
Mean value (normalized dist.)	$\langle x \rangle = \sum_{i=1}^n x_i p_i$	$\langle x \rangle = \int_{x_{\min}}^{x_{\max}} x f(x) dx$
Expected value (normalized dist.)	$\langle g \rangle = \sum_{i=1}^n g(x_i) p_i$	$\langle g \rangle = \int_{x_{\min}}^{x_{\max}} g(x) f(x) dx$

Examples and Applications

- Examples

- Binomial probability distribution: discrete
- Uniform probability distribution: continuous
- Gaussian probability distribution: continuous

- Applications

- Quantum mechanics
- Kinetic theory of a gas

Binomial Probability Distribution

- p = probability that “heads” will occur
- $(1 - p)$ = probability that “tails” will occur

$$P(n, m) = \frac{n!}{m! (n - m)!} p^m (1 - p)^{n-m}$$

Mean Value and Variance

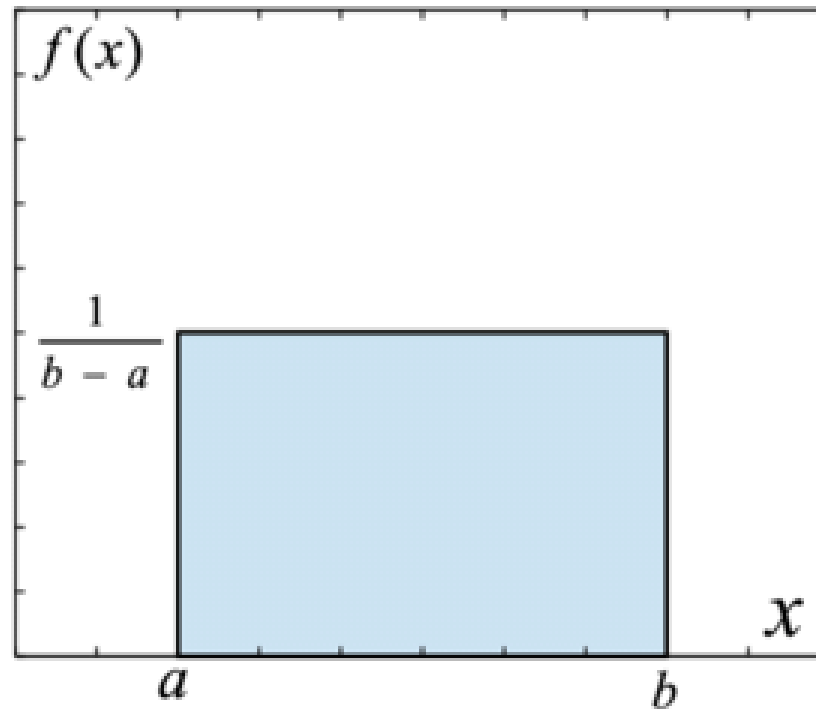
- Mean value

$$\langle x \rangle = \sum_{i=1}^{\infty} p_i x_i = \sum_{m=0}^n \frac{n!}{m! (n-m)!} p^m (1-p)^{n-m} \times m$$
$$\langle x \rangle = np$$

- Variance

$$\sigma_x^2 = \langle x^2 \rangle - \langle x \rangle^2 = np(1-p)$$

Uniform Probability Distribution



Mean Value and Variance

- Mean value

$$\langle x \rangle = \frac{a + b}{2}$$

- Variance

$$\sigma_x^2 = \frac{(a - b)^2}{12}$$

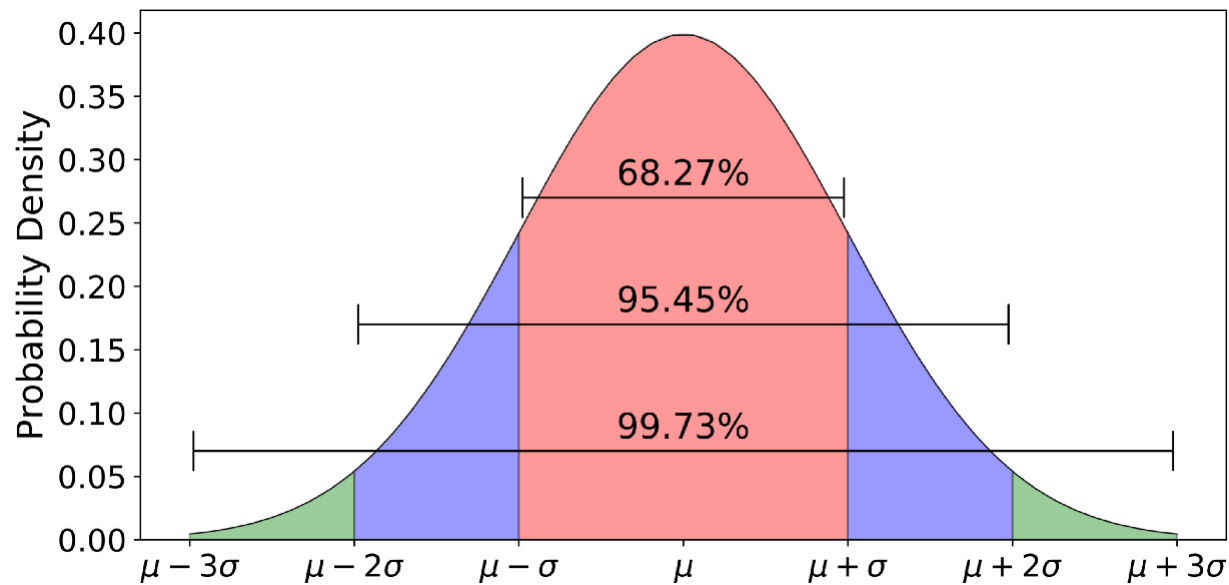
Today's Class

- Gaussian probability distribution
- Applications
 - Quantum mechanics
 - Kinetic theory of a gas
- Sample statistics
- Confidence interval determination



Gaussian Probability Distribution

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left[-\frac{(x - \mu)^2}{2\sigma^2} \right]$$



Example 15.6

Show that the Gaussian distribution satisfies the normalization condition.

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(x - \mu)^2}{2\sigma^2}\right]$$

The normalization condition states

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

Letting $t = (x - \mu)/\sqrt{2\sigma^2}$,

$$\int_{-\infty}^{\infty} f(x) dx = \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{\infty} e^{-t^2} (\sqrt{2\sigma^2} dt) = \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-t^2} dt = 1$$

Hence, the Gaussian distribution satisfies the normalization condition.

Mean Value and Variance

- Mean value

$$\langle x \rangle = \int_{-\infty}^{\infty} x f(x) dx = \mu$$

- Variance

$$\sigma_x^2 = \langle x^2 \rangle - \langle x \rangle^2 = \sigma^2$$

Error Function (오차함수)

$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$$

(Fraction between $\mu - \alpha$ and $\mu + \alpha$)

$$\begin{aligned} &= \int_{\mu-\alpha}^{\mu+\alpha} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/2\sigma^2} dx = \int_{-\alpha/\sqrt{2\sigma^2}}^{\alpha/\sqrt{2\sigma^2}} \frac{1}{\sqrt{\pi}} e^{-t^2} dt \\ &= \frac{2}{\sqrt{\pi}} \int_0^{\alpha/\sqrt{2\sigma^2}} e^{-t^2} dt = \operatorname{erf}\left(\frac{\alpha}{\sqrt{2\sigma^2}}\right) \end{aligned}$$

Appendix G

Values of the Error Function

x		0	1	2	3	4	5	6	7	8	9
0.0	0.0	000	113	226	338	451	564	676	789	901	013*
0.1	0.1	125	236	348	459	569	680	790	900	009*	118*
0.2	0.2	227	335	443	550	657	763	869	974	079*	183*
0.3	0.3	286	389	491	593	694	794	893	992	090*	187*
0.4	0.4	284	380	475	569	662	755	847	937	027*	117*
0.5	0.5	205	292	379	465	549	633	716	798	879	959
0.6	0.6	039	117	194	270	346	420	494	566	638	708
0.7		778	847	914	981	047*	112*	175*	238*	300*	361*
0.8	0.7	421	480	538	595	651	707	761	814	867	918
0.9		969	019*	068*	116*	163*	209*	254*	299*	342*	385*
1.0	0.8	427	468	508	548	586	624	661	698	733	768
1.1		802	835	868	900	931	961	991	020*	048*	076*
1.2	0.9	103	130	155	181	205	229	252	275	297	319
1.3		340	361	381	400	419	438	456	473	490	507
1.4	0.95	23	39	54	69	83	97	11*	24*	37*	49*
1.5	0.96	61	73	84	95	06*	16*	26*	36*	45*	55*
1.6	0.97	63	72	80	88	96	04*	11*	18*	25*	32*
1.7	0.98	38	44	50	56	61	67	72	77	82	86
1.8		91	95	99	03*	07*	11*	15*	18*	22*	25*
1.9	0.99	28	31	34	37	39	42	44	47	49	51
2.0	0.995	32	52	72	91	09*	26*	42*	58*	73*	88*
2.1	0.997	02	15	28	41	53	64	75	85	95	05*
2.2	0.998	14	22	31	39	46	54	61	67	74	80
2.3		86	91	97	02*	06*	11*	15*	20*	24*	28*
2.4	0.999	31	35	38	41	44	47	50	52	55	57
2.5		59	61	63	65	67	69	71	72	74	75
2.6		76	78	79	80	81	82	83	84	85	86
2.7		87	87	88	89	89	90	91	91	92	92
2.8	0.9999	25	29	33	37	41	44	48	51	54	56
2.9		59	61	64	66	68	70	72	73	75	77

* From Eugene Jahnke and Fritz Emde, *Tables of Functions*, Dover Publications, New York, 1945, p. 24.



Example 15.7

Find the probability that x lies between $\mu - 0.5\sigma$ and $\mu + 0.5\sigma$ for a Gaussian distribution.

$$(\text{Fraction between } \mu - \alpha \text{ and } \mu + \alpha) = \text{erf}\left(\frac{\alpha}{\sqrt{2}\sigma}\right)$$

Thus,

$$\text{Probability} = \text{erf}\left(\frac{0.5\sigma}{\sqrt{2}\sigma}\right) = \text{erf}\left(\frac{1}{2\sqrt{2}}\right) = \text{erf}(0.35) = 0.38$$

Central Limit Theorem (중심극한정리)

For random variables $x_1, x_2, \dots, x_n$,

$$y = x_1 + x_2 + \dots + x_n$$

is governed by a probability distribution that approaches
a Gaussian distribution for large n

- If experimental errors arise from multiples sources, they should be at least approximately described by a Gaussian distribution

Application: Quantum Mechanics

For a wave function ψ of the system,

$$\text{Probability distribution} \propto |\psi|^2$$

e.g. Ground state of a 1-dimensional particle in a box

- Wave function $\psi(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{\pi x}{L}\right)$
- Probability distribution $f(x) = |\psi|^2 = \frac{2}{L} \sin^2\left(\frac{\pi x}{L}\right)$

Modified Example 15.9

For a particle in a box of length L in its ground state, find the probability that the particle will be found in $(0, L/2)$.

$$\text{Probability} = \int_0^{L/2} f(x) \, dx = \int_0^{L/2} |\psi(x)|^2 \, dx$$

$$\text{Since } f(x) = |\psi|^2 = \frac{2}{L} \sin^2 \left(\frac{\pi x}{L} \right),$$

$$\text{Probability} = \int_0^{L/2} \frac{2}{L} \sin^2 \left(\frac{\pi x}{L} \right) \, dx = \frac{2}{\pi} \int_0^{\pi/2} \sin^2 t \, dt = \frac{1}{2}$$

Expectation Value (기댓값)

“The predicted mean value of
many measurements of an observable”

For a wave function ψ of the system,

$$\langle A \rangle = \int \psi^* \hat{A} \psi \, dq$$

where $\hat{A}$ is the operator corresponding to the variable A .

Example 15.8

Find the expressions for $\langle x \rangle$ and σ_x for a particle in a box of length L in its ground state.

$$\psi(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{\pi x}{L}\right)$$

The expectation values are

$$\langle x \rangle = \int_0^L \psi^*(x) \hat{x} \psi(x) dx = \int_0^L x |\psi|^2 dx = \frac{2}{L} \int_0^L x \sin^2\left(\frac{\pi x}{L}\right) dx = \frac{L}{2}$$

$$\langle x^2 \rangle = \int_0^L \psi^*(x) \hat{x}^2 \psi(x) dx = \int_0^L x^2 |\psi|^2 dx = \frac{2}{L} \int_0^L x^2 \sin^2\left(\frac{\pi x}{L}\right) dx = \frac{L^2}{3}$$

$$\sigma_x = \sqrt{\sigma_x^2} = \sqrt{\langle x^2 \rangle - \langle x \rangle^2} = \sqrt{\frac{L^2}{3} - \frac{L^2}{4}} = \frac{L}{\sqrt{12}}$$

1-Dimensional Hamiltonian Operator

$$\hat{H} = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + \mathcal{V}(x)$$

- Corresponds to the system energy
- For a particle in a box, $\mathcal{V}(x) = 0$

Example 15.10

Find the expectation value of the energy and its standard deviation for the ground state of our particle in a box.

$$\psi(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{\pi x}{L}\right); \quad \hat{H} = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2}$$

$$\begin{aligned} \langle E \rangle &= \int_0^L \psi^*(x) \hat{H} \psi(x) dx = \int_0^L \sqrt{\frac{2}{L}} \sin\left(\frac{\pi x}{L}\right) \left[-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \left\{ \sqrt{\frac{2}{L}} \sin\left(\frac{\pi x}{L}\right) \right\} \right] dx \\ &= -\frac{\hbar^2}{mL} \int_0^L \sin\left(\frac{\pi x}{L}\right) \left(-\frac{\pi^2}{L^2} \right) \sin\left(\frac{\pi x}{L}\right) dx = \frac{\pi^2 \hbar^2}{mL^3} \int_0^L \sin^2\left(\frac{\pi x}{L}\right) dx \\ &= \frac{\pi^2 \hbar^2}{mL^3} \times \frac{L}{2} = \frac{\pi^2 \hbar^2}{2mL^2} = \frac{h^2}{8mL^2} \end{aligned}$$

Example 15.10

Find the expectation value of the energy and its standard deviation for the ground state of our particle in a box.

$$\psi(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{\pi x}{L}\right); \quad \hat{H} = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2}$$

$$\begin{aligned} \langle E^2 \rangle &= \int_0^L \psi^*(x) \hat{H}^2 \psi(x) dx = \int_0^L \sqrt{\frac{2}{L}} \sin\left(\frac{\pi x}{L}\right) \left[\left(\frac{\hbar^2}{2m}\right)^2 \frac{d^4}{dx^4} \left\{ \sqrt{\frac{2}{L}} \sin\left(\frac{\pi x}{L}\right) \right\} \right] dx \\ &= \frac{\hbar^4}{2m^2 L} \int_0^L \sin\left(\frac{\pi x}{L}\right) \left(\frac{\pi^4}{L^4}\right) \sin\left(\frac{\pi x}{L}\right) dx = \frac{\pi^4 \hbar^4}{2m^2 L^5} \int_0^L \sin^2\left(\frac{\pi x}{L}\right) dx \\ &= \frac{\pi^4 \hbar^4}{2m^2 L^5} \times \frac{L}{2} = \frac{\pi^4 \hbar^4}{4m^2 L^4} = \frac{h^4}{64m^2 L^4} \end{aligned}$$

Example 15.10

Find the expectation value of the energy and its standard deviation for the ground state of our particle in a box.

$$\langle E \rangle = \frac{h^2}{8mL^2}; \quad \langle E^2 \rangle = \frac{h^4}{64m^2L^4}$$

The standard deviation is

$$\sigma_E = \sqrt{\sigma_E^2} = \sqrt{\langle E^2 \rangle - \langle E \rangle^2} = \sqrt{\frac{h^4}{64m^2L^4} - \left(\frac{h^2}{8mL^2}\right)^2} = 0$$

Application: Kinetic Theory of a Gas

- **Boltzmann factor:** in the thermal equilibrium of temperature T , the probability of having energy E is

$$f(E) \propto e^{-E/k_B T}$$

- The probability that a gas molecule has the velocity (v_x, v_y, v_z) is

$$f(v_x, v_y, v_z) \propto \exp\left(-\frac{m(v_x^2 + v_y^2 + v_z^2)}{2k_B T}\right)$$

Normalization

$$f(v_x, v_y, v_z) = A \exp\left(-\frac{m(v_x^2 + v_y^2 + v_z^2)}{2k_B T}\right)$$

$$I = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(v_x, v_y, v_z) dv_x dv_y dv_z = 1$$

$$I = A \int_{-\infty}^{\infty} e^{-mv_x^2/2k_B T} dv_x \int_{-\infty}^{\infty} e^{-mv_y^2/2k_B T} dv_y \int_{-\infty}^{\infty} e^{-mv_z^2/2k_B T} dv_z$$

$$I = A \left(\sqrt{\frac{2\pi k_B T}{m}} \right)^3 \rightarrow A = \left(\frac{m}{2\pi k_B T} \right)^{3/2}$$

Velocity to Speed

- $f(v_x, v_y, v_z) dv_x dv_y dv_z$:

Probability that a molecule has a velocity in the range

$$[v_x, v_x + dv_x], [v_y, v_y + dv_y], [v_z, v_z + dv_z]$$

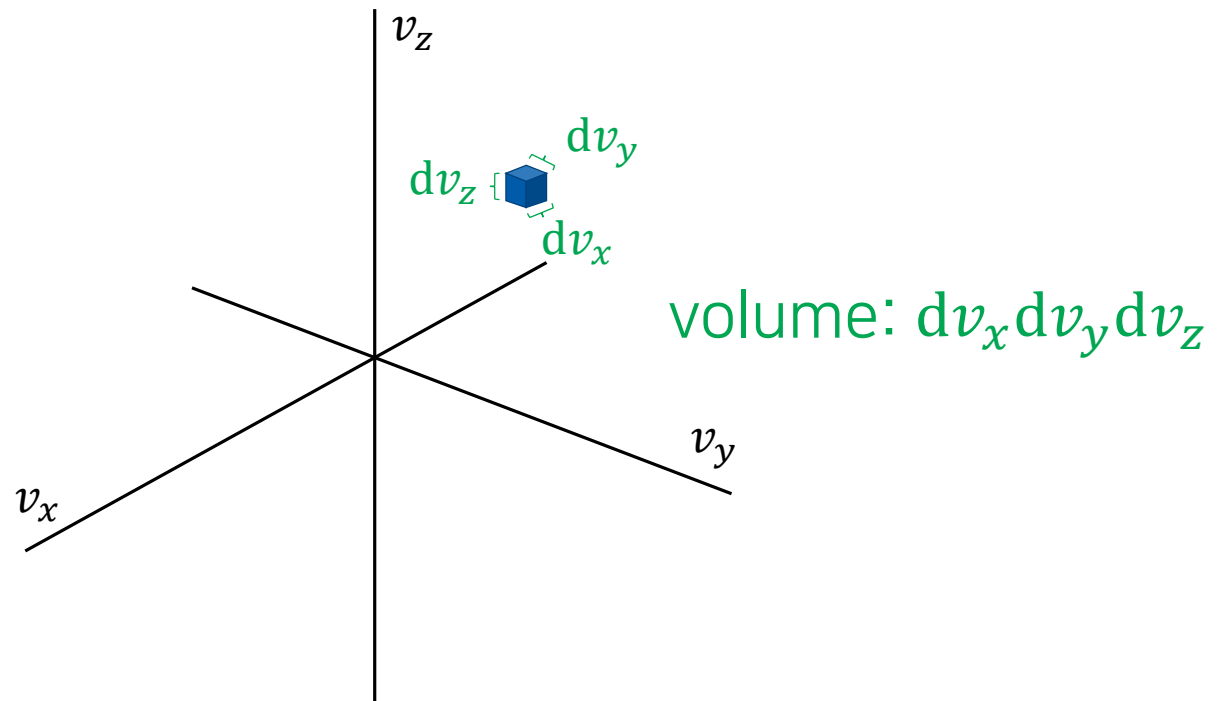
- $f_v(v) dv$:

Probability that a molecule has a speed in the range

$$[v, v + dv]$$

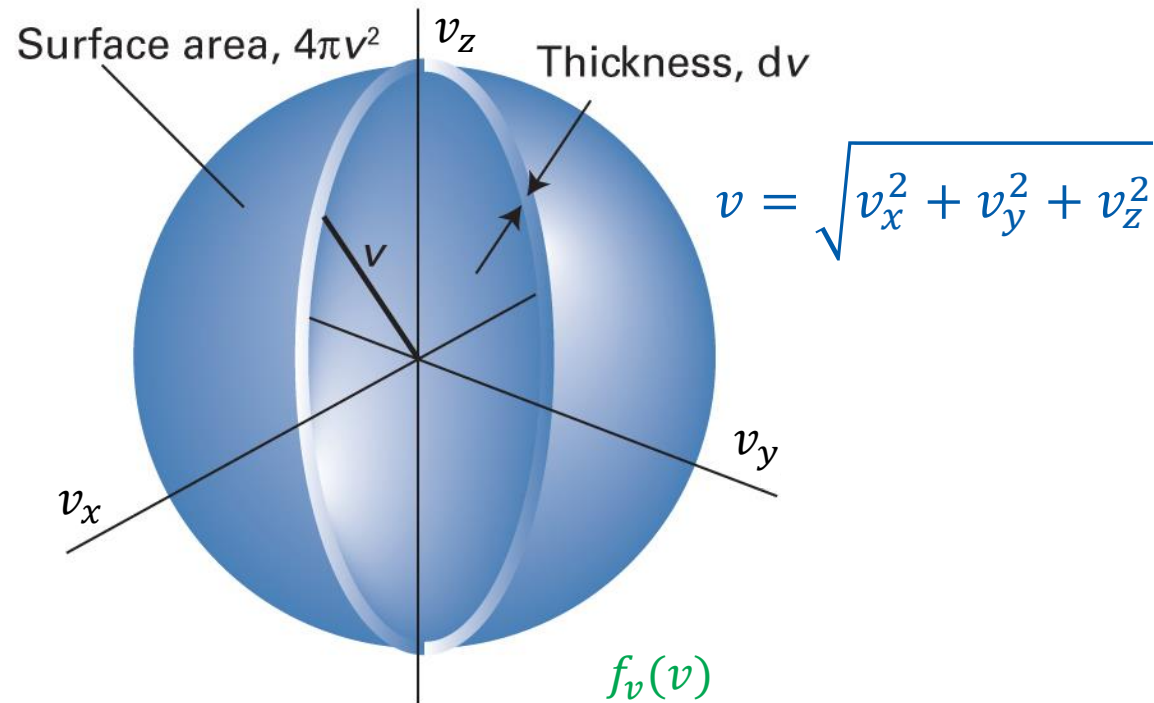
Can we obtain $f_v(v)dv$ from $f(v_x, v_y, v_z)dv_x dv_y dv_z$?

Velocity Space



$$f(v_x, v_y, v_z) dv_x dv_y dv_z = \text{function} \times \text{volume}$$

Velocity Space



$$\text{function} \times \text{volume} = \overbrace{f(v_x, v_y, v_z)}^{f_v(v)} \times 4\pi v^2 dv$$

(Image: Atkins *et al.*, *Atkins' Physical Chemistry*, 11th ed., Figure 1B.3)

Maxwell-Boltzmann Distribution

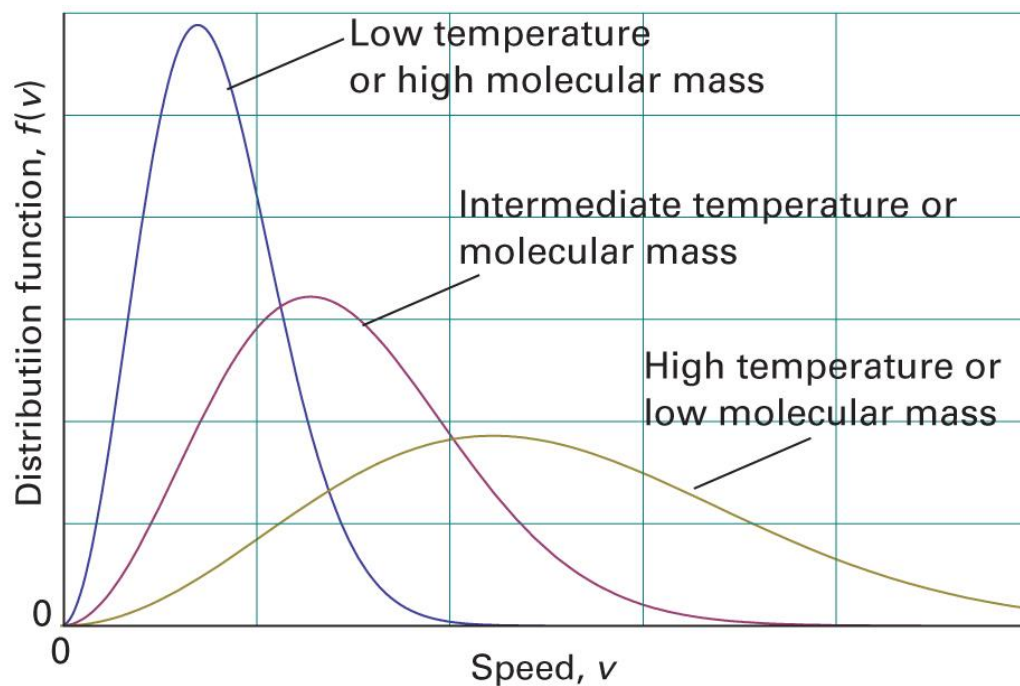
$$f_v(v) = f(v_x, v_y, v_z) 4\pi v^2$$

$$f_v(v) = 4\pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} v^2 e^{-m(v_x^2 + v_y^2 + v_z^2)/2k_B T}$$

$$f_v(v) = 4\pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} v^2 e^{-mv^2/2k_B T}$$

Behaviors of the Distribution

$$f_v(v) = 4\pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} v^2 e^{-mv^2/2k_B T}$$



(Image: Atkins *et al.*, *Atkins' Physical Chemistry*, 11th ed., Figure 1B.4)

Mean Speed (평균 속력)

$$f_v(v) = 4\pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} v^2 e^{-mv^2/2k_B T}$$

$$\langle v \rangle = \int_0^\infty v f_v(v) dv = \int_0^\infty 4\pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} v^3 e^{-mv^2/2k_B T} dv$$

$$= 4\pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} \int_0^\infty v^3 e^{-mv^2/2k_B T} dv$$

$$= 4\pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} \frac{(2k_B T)^2}{2m^2} = \left(\frac{8k_B T}{\pi m} \right)^{1/2}$$

Example 15.12

Find a formula for the mean of the square of the speed and the root-mean-square speed of molecules in a gas.

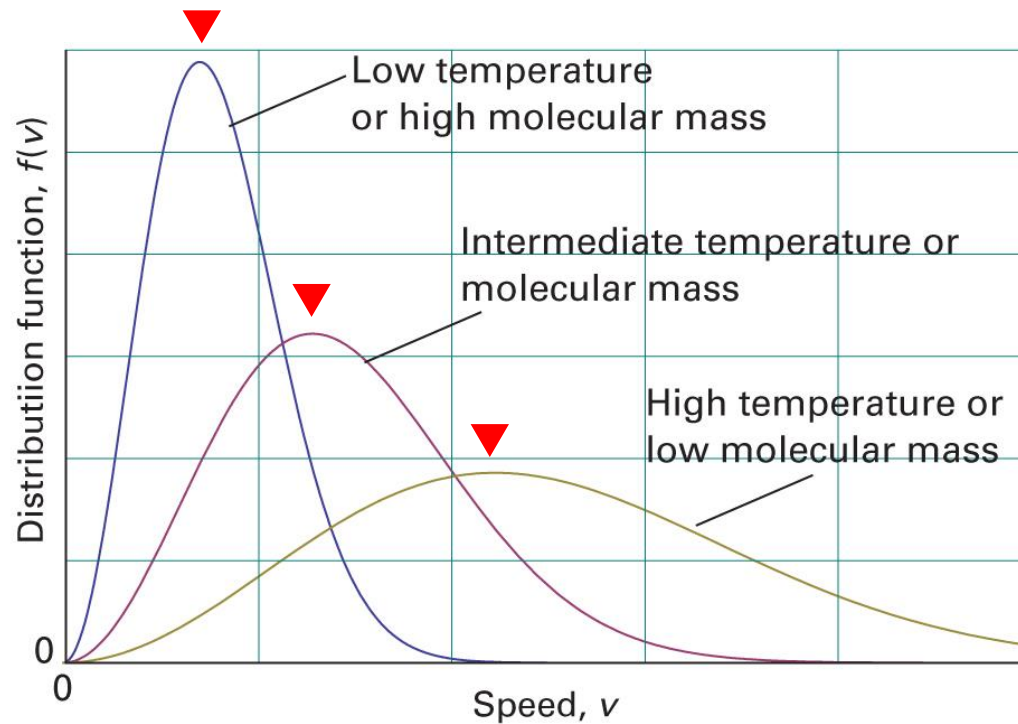
$$f_v(v) = 4\pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} v^2 e^{-mv^2/2k_B T}$$

$$\langle v^2 \rangle = \int_0^\infty v^2 f_v(v) dv = \int_0^\infty 4\pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} v^4 e^{-mv^2/2k_B T} dv$$

$$= 4\pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} \int_0^\infty v^4 e^{-mv^2/2k_B T} dv = \frac{3k_B T}{m}$$

$$v_{\text{rms}} = \sqrt{\langle v^2 \rangle} = \left(\frac{3k_B T}{m} \right)^{1/2}$$

Most Probable Speed (최빈 속력)



$$\frac{df(v)}{dv} = 0$$

(Image: Atkins *et al.*, *Atkins' Physical Chemistry*, 11th ed., Figure 1B.4)

Most Probable Speed (최빈 속력)

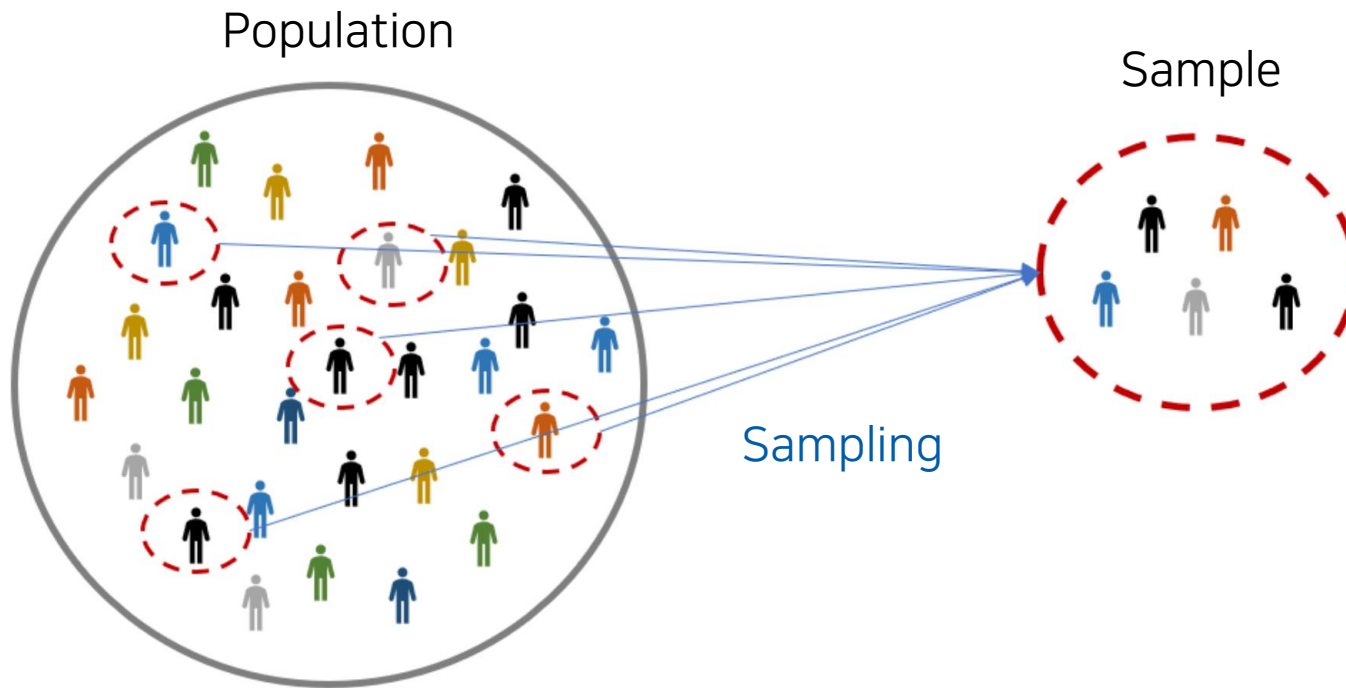
$$f_v(v) = 4\pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} v^2 e^{-mv^2/2k_B T}$$

$$\frac{df_v(v)}{dv} = 4\pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} \left(2v e^{-mv^2/2k_B T} - \frac{mv^3}{k_B T} e^{-mv^2/2k_B T} \right) = 0$$

$$2v - \frac{mv^3}{k_B T} = 0$$

$$v_{\text{mp}} = \left(\frac{2k_B T}{m} \right)^{1/2}$$

Population (모집단) and Sample (표본)



(Image: <https://www.omniconvert.com/what-is/sample-size/>)

Statistics of a Sample

- Goal: estimate mean, variance, standard deviation of the population
 - After an infinite number of samplings,
 - $\langle \text{sample mean} \rangle \rightarrow \text{population mean}$
 - $\langle \text{sample variance} \rangle \rightarrow \text{population variance}$
 - $\langle \text{sample standard deviation} \rangle$
 $\rightarrow \text{population standard deviation}$
- (unbiased estimates)

Sample Mean

For a sample of N measurements, the sample mean is

$$\langle x \rangle = \frac{1}{N} \sum_{i=1}^N x_i$$

(same formula with the population mean)

Sample Variance

For a sample of N measurements, the sample variance is

$$s_x^2 = \frac{1}{N-1} \sum_{i=1}^N (x_i - \langle x \rangle)^2$$

- Population variance (population size = M)

$$\sigma_x^2 = \frac{1}{M} \sum_{i=1}^M (x_i - \langle x \rangle)^2$$

Sample Standard Deviation

For a sample of N measurements, the sample standard deviation is

$$s_x = \left[\frac{1}{N-1} \sum_{i=1}^N (x_i - \langle x \rangle)^2 \right]^{1/2} = \sqrt{s_x^2}$$

Example 15.14

Find the mean and the standard deviation of the sample of numbers
32.41, 33.76, 32.91, 33.04, 32.75, 33.23.

$$\langle x \rangle = \frac{1}{N} \sum_{i=1}^N x_i = \frac{1}{6} (32.41 + 33.76 + 32.91 + 33.04 + 32.75 + 33.23) = 33.02$$

$$\begin{aligned} s_x^2 &= \frac{1}{N-1} \sum_{i=1}^N (x_i - \langle x \rangle)^2 \\ &= \frac{1}{5} [(32.41 - 33.02)^2 + (33.76 - 33.02)^2 + (32.91 - 33.02)^2 \\ &\quad + (33.04 - 33.02)^2 + (32.75 - 33.02)^2 + (33.23 - 33.02)^2] = 0.21 \end{aligned}$$


$$s_x = \sqrt{s_x^2} = 0.46$$

Confidence Level (신뢰수준)

- 95% confidence level = errors have a 95% probability of being correct
- If we want to assert that the interval

$$\langle x \rangle - \varepsilon < x < \langle x \rangle + \varepsilon$$

has a 95% chance of containing the true value x_{true} , the interval is called the **95% confidence interval** (95% 신뢰 구간) and ε the **estimated error** (추정 오차)

How to Determine ε

Under some mathematical assumptions, we can derive an estimate for ε :

$$\varepsilon = t(N - 1, \alpha) \frac{s_x}{\sqrt{N}}$$

- N : sample size
- α : significance ($1 - \alpha$ = confidence level)
- $t(N - 1, \alpha)$: Student t value

Table 15.1

$(v = N - 1)$

Number of degrees of freedom v	Maximum value of Student's t factor for the significance levels indicated			
	$t(v, 0.60)$	$t(v, 0.10)$	$t(v, 0.05)$	$t(v, 0.01)$
1	1.376	6.314	12.706	63.657
2	1.061	2.920	4.303	9.925
3	0.978	2.353	3.182	5.841
4	0.941	2.132	2.776	4.604
5	0.920	2.015	2.571	4.032
6	0.906	1.943	2.447	3.707
7	0.896	1.895	2.365	3.499
8	0.889	1.860	2.306	3.355
9	0.883	1.833	2.262	3.250
10	0.879	1.812	2.228	3.169
11	0.876	1.796	2.201	3.106
12	0.873	1.782	2.179	3.055
13	0.870	1.771	2.160	3.012
14	0.868	1.761	2.145	2.977
15	0.866	1.753	2.131	2.947
16	0.865	1.746	2.120	2.921
17	0.863	1.740	2.110	2.898
18	0.862	1.734	2.101	2.878
19	0.861	1.729	2.093	2.861
20	0.860	1.725	2.086	2.845
21	0.859	1.721	2.080	2.831
22	0.858	1.717	2.074	2.819
23	0.858	1.714	2.069	2.807
24	0.857	1.711	2.064	2.797
25	0.856	1.708	2.060	2.787
26	0.856	1.706	2.056	2.479
27	0.855	1.703	2.052	2.771
28	0.855	1.701	2.048	2.763
29	0.854	1.699	2.045	2.756
30	0.854	1.697	2.042	2.750
40	0.851	1.684	2.021	2.704
60	0.848	1.671	2.000	2.660
∞	0.842	1.645	1.960	2.576

Modified Example 15.15

Determine the 95% interval for the set of measured temperatures

42.70°C, 42.60°C, 42.78°C, 42.83°C.

$$\langle x \rangle - \varepsilon < x < \langle x \rangle + \varepsilon; \quad \varepsilon = t(N - 1, \alpha) \frac{s_x}{\sqrt{N}}$$

$$\langle x \rangle = 42.73^\circ\text{C}$$

$$s_x^2 = 0.0303^\circ\text{C}^2$$

$$s_x = 0.17^\circ\text{C}$$

$$\varepsilon = t(3, 0.05) \frac{s_x}{\sqrt{N}} = 3.182 \times \frac{0.17^\circ\text{C}}{\sqrt{4}} = 0.27^\circ\text{C}$$

Hence, at the 95% confidence level, $x = 42.73^\circ\text{C} \pm 0.27^\circ\text{C}$